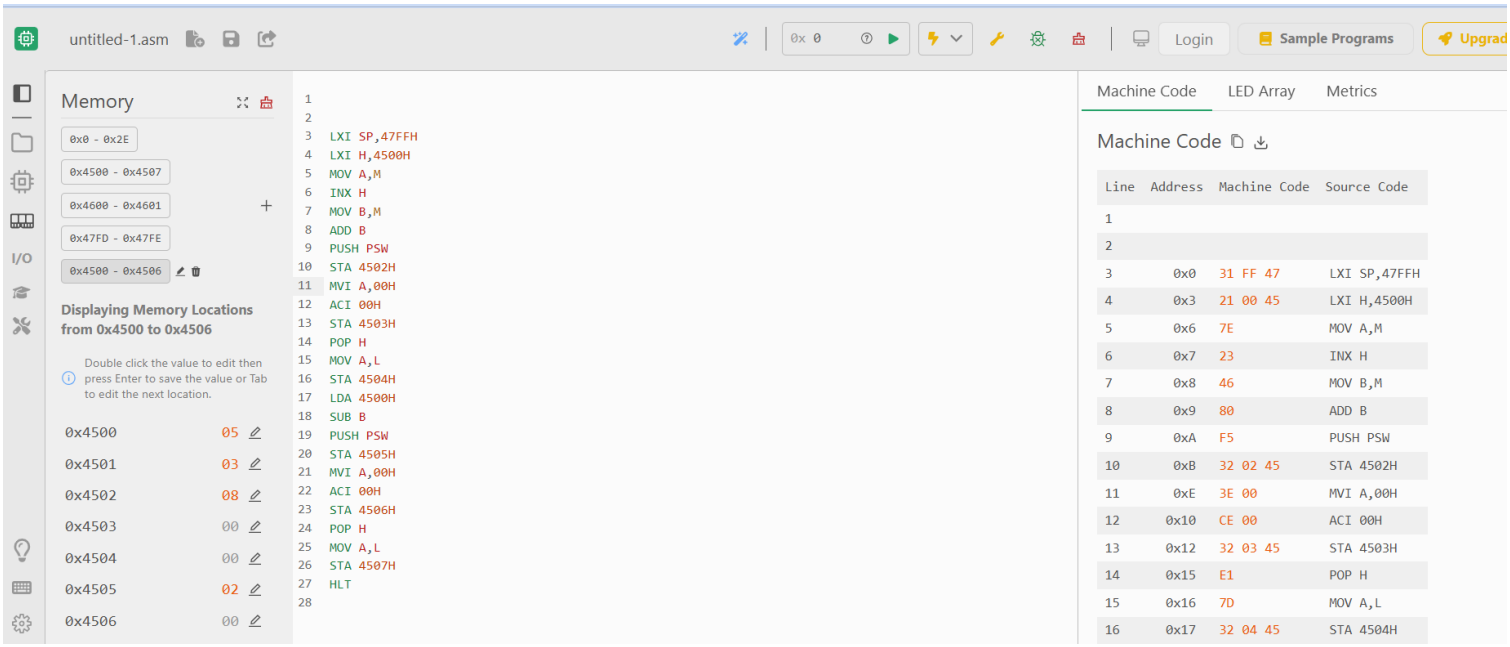
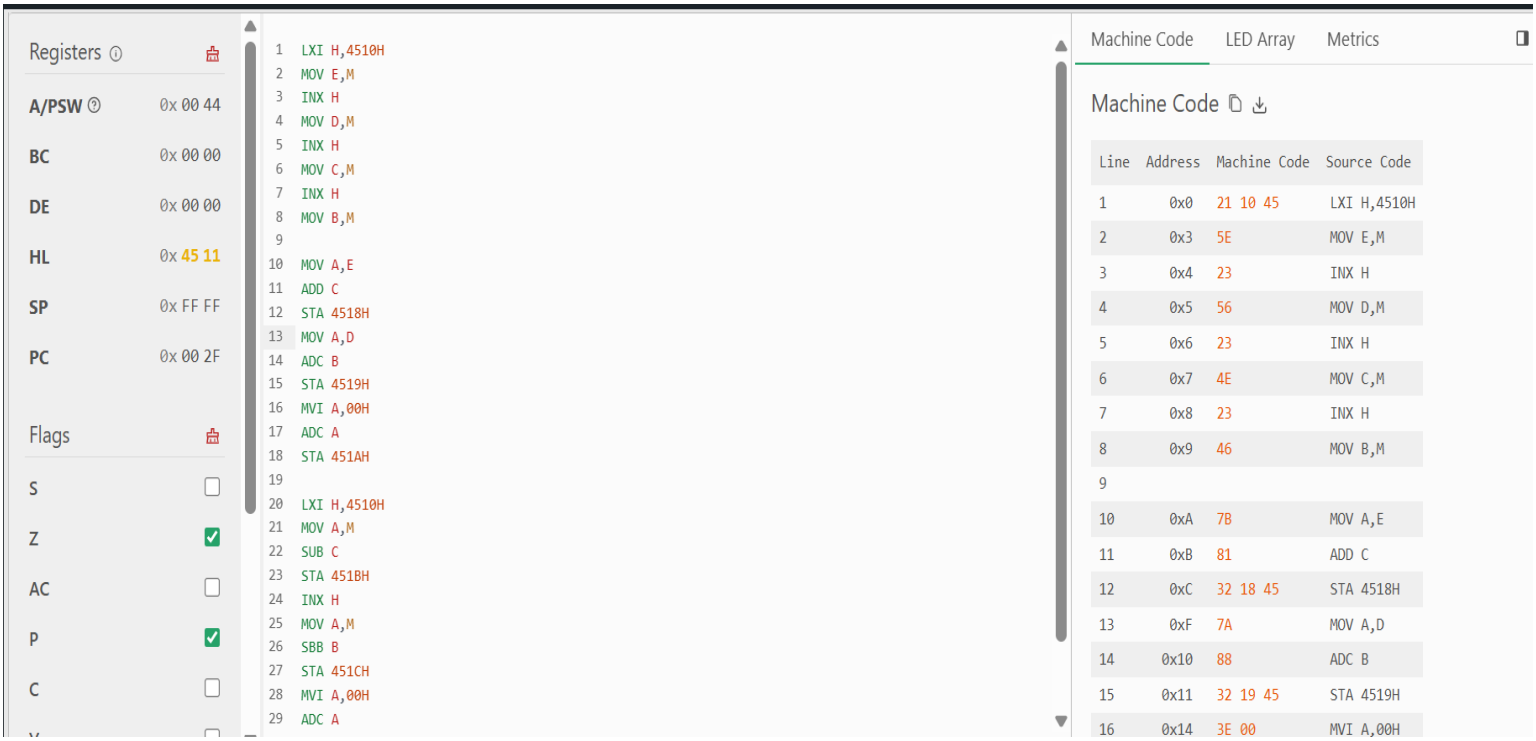
a. EXPERIMENT 1

PROGRAM A :



PROGRAM B:



Displaying Memory Locations from 0x4510 to 0x451D

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4510	34
0x4511	12
0x4512	78
0x4513	56
0x4514	00
0x4515	00
0x4516	00
0x4517	00
0x4518	AC
0x4519	68
0x451A	00
0x451B	BC
0x451C	BB
0x451D	01

```
1  LXI H,4510H
2  MOV E,M
3  INX H
4  MOV D,M
5  INX H
6  MOV C,M
7  INX H
8  MOV B,M
9
10 MOV A,E
11 ADD C
12 STA 4518H
13 MOV A,D
14 ADC B
15 STA 4519H
16 MVI A,00H
17 ADC A
18 STA 451AH
19
20 LXI H,4510H
21 MOV A,M
22 SUB C
23 STA 451BH
24 INX H
25 MOV A,M
26 SBB B
27 STA 451CH
28 MVI A,00H
29 ADC A
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	21 10 45	LXI H,4510H
2	0x3	5E	MOV E,M
3	0x4	23	INX H
4	0x5	56	MOV D,M
5	0x6	23	INX H
6	0x7	4E	MOV C,M
7	0x8	23	INX H
8	0x9	46	MOV B,M
9			
10	0xA	7B	MOV A,E
11	0xB	81	ADD C
12	0xC	32 18 45	STA 4518H
13	0xF	7A	MOV A,D
14	0x10	88	ADC B
15	0x11	32 19 45	STA 4519H
16	0x14	3E 00	MVI A,00H

PROGRAM D :

Registers

A/PSW	0x FF A5
BC	0x 00 04
DE	0x 00 04
HL	0x 00 04
SP	0x 47 FF
PC	0x 00 4E

Flags

S	☑
Z	☐
AC	☐
P	☑
C	☑
V	☐

```
1  LXI SP,47FFH
2  LHLD 4530H
3  XCHG
4  LHLD 4532H
5  MOV B,H
6  MOV C,L
7  LXI H,0000H
8  SHLD 4536H
9  MOV A,B
10 ORA C
11 JZ STORE
12 MLOOP: DAD D
13 JNC NOCY
14 PUSH H
15 LHLD 4536H
16 INX H
17 SHLD 4536H
18 POP H
19 NOCY: DCX B
20 MOV A,B
21 ORA C
22 JNZ MLOOP
23 STORE: SHLD 4534H
24 LHLD 453AH
25 XCHG
26 LHLD 4538H
27 LXI B,0000H
28 DLOOP: MOV A,L
29 SUB E
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	2A 30 45	LHLD 4530H
3	0x6	EB	XCHG
4	0x7	2A 32 45	LHLD 4532H
5	0xA	44	MOV B,H
6	0xB	4D	MOV C,L
7	0xC	21 00 00	LXI H,0000H
8	0xF	22 36 45	SHLD 4536H
9	0x12	78	MOV A,B
10	0x13	B1	ORA C
11	0x14	CA 2A 00	JZ STORE
12	0x17	19	MLOOP: DAD D
13	0x18	D2 24 00	JNC NOCY
14	0x1B	E5	PUSH H
15	0x1C	2A 36 45	LHLD 4536H
16	0x1F	23	INX H

Displaying Memory Locations from 0x4530 to 0x453F

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4530	05
0x4531	00
0x4532	03
0x4533	00
0x4534	0F
0x4535	00
0x4536	00
0x4537	00
0x4538	11
0x4539	00
0x453A	04
0x453B	00
0x453C	04
0x453D	00

```
1  LXI SP,47FFH
2  LHLD 4530H
3  XCHG
4  LHLD 4532H
5  MOV B,H
6  MOV C,L
7  LXI H,0000H
8  SHLD 4536H
9  MOV A,B
10 ORA C
11 JZ STORE
12 MLOOP: DAD D
13 JNC NOCY
14 PUSH H
15 LHLD 4536H
16 INX H
17 SHLD 4536H
18 POP H
19 NOCY: DCX B
20 MOV A,B
21 ORA C
22 JNZ MLOOP
23 STORE: SHLD 4534H
24 LHLD 453AH
25 XCHG
26 LHLD 4538H
27 LXI B,0000H
28 DLOOP: MOV A,L
29 SUB E
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	2A 30 45	LHLD 4530H
3	0x6	EB	XCHG
4	0x7	2A 32 45	LHLD 4532H
5	0xA	44	MOV B,H
6	0xB	4D	MOV C,L
7	0xC	21 00 00	LXI H,0000H
8	0xF	22 36 45	SHLD 4536H
9	0x12	78	MOV A,B
10	0x13	B1	ORA C
11	0x14	CA 2A 00	JZ STORE
12	0x17	19	MLOOP: DAD D
13	0x18	D2 24 00	JNC NOCY
14	0x1B	E5	PUSH H
15	0x1C	2A 36 45	LHLD 4536H
16	0x1F	23	INX H

PROGRAM A :

GIVEN:

4500H = 05H

4501H = 03H

Addition (ADD B)

A = 05H = 0000 0101

B = 03H = 0000 0011

```
  0000  0101
  0000  0011
  ----  ----
  0000  1000
```

PROGRAM B :

Given X = 1234H

Y = 5678H

(i) ADDITION :

```
  1234H     Sum = 68ACH
  5678H     Carry = 00H
  ----
  68ACH
```

(ii) SUBTRACTION :

```
  1234H
- 5678H
  ----
  BBBCH   (2's complement)
```

Difference : BB BCH

Borrow = 01H

PROGRAM D :

(i) Multiplication (X x Y)

Given :

X = 0005H = 5 ; Y = 0003H = 3

Calculation :

X x Y = 5 x 3 = 15 Product = 000F H

15_{10} = 000F H

(ii) DIVISION :

Given :

Dividend = 0011H = 17

Divisor = 0004H = 4

17 ÷ 4 = 4 remainder 1

17 = (4 x 4) + 1

= 16 + 1

Quotient = 0004H

Remainder = 0001H